

TRIP

Tanzania Readmission Intelligence Platform

PERSONAS · ROLES · DASHBOARDS · SYSTEM COLLABORATION

National → Community | 35 Roles | Tanzania Health Stack Integration | March 2026

35 System Roles	7 Hierarchy Tiers	15 System Integrations	AI Injection Model
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1. Design Philosophy — The AI Injection Model

TRIP is not a standalone health system. It is an intelligence layer injected into Tanzania's existing health stack. The distinction matters institutionally: systems like GoT-HoMIS, Afya eHMS, and DHIS2 took years to build and billions of shillings to deploy. TRIP does not compete with them — it makes them smarter. This is the same model that Epic uses with its Predictive Risk models (sepsis, readmission, deterioration) and that Cerner used with HealthIntent — neither replaced the EHR; both added intelligence on top of it.

The core product promise is simple: at the moment a clinician is about to discharge a patient, TRIP tells them the probability that patient will be back within 30 days and why, in plain language, so they can act on it before the patient leaves the building. Every role in the system exists to support, govern, or improve that moment of intelligence delivery.

>> Every dashboard, every role, every integration in this document exists to serve one clinical moment: a clinician at discharge, holding a risk score they trust, with an intervention plan already generated, and a CHW already notified to follow up. Everything else is governance and scale.

1.1 How This Compares to Global Benchmarks

Platform	Intelligence Approach	Tanzania Adaptation
Epic (US)	Embedded ML risk scores inside EHR discharge workflow; CDS Hooks standard; real-time alerts	TRIP embeds in Afya eHMS and GoT-HoMIS discharge screens via FHIR CDS Hooks — same pattern
Cerner / Oracle Health	HealthIntent population health: risk stratification across patient populations, not just at discharge	TRIP's RHMT and MOH dashboards replicate this: population-level readmission burden by region/district
Singapore NEHR	National electronic health record: single patient identity across all facilities; unified clinical history	TRIP uses pseudonym+DOB+phone probabilistic matching as UPI proxy until Tanzania UPI is deployed
India ABDM	Federated health record: consent-based data sharing between facility systems; FHIR R4 standard	TRIP uses same OpenHIE/FHIR pattern: facility systems push data to TRIP's central intelligence layer
Rwanda RBC Health	District-level integrated health management: offline-first, low-bandwidth, CHW-integrated	TRIP's offline PWA, CHW follow-up module, and Africa's Talking SMS are direct adaptations of Rwanda's model
Kenya KHIS / OpenMRS	DHIS2 + OpenMRS integration via FHIR R4 mediator as standard data bridge	TRIP uses exact same mediator architecture: GoT-HoMIS -> FHIR2 module -> TRIP data lake

2. Full Role Taxonomy — 35 Roles Across 7 Tiers

Tanzania's Ministry of Health structure runs from the national cabinet level to the individual village health worker. The following taxonomy maps every human role that touches TRIP — either as a data producer, intelligence consumer, governance authority, or system operator. Roles are grouped by hierarchy tier.

TIER 1 — NATIONAL (Ministry of Health)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
MOH_PS	Permanent Secretary, MoH	National aggregate read-only	Strategic oversight; MoH annual health report consumer
MOH_DCS	Director of Curative Services	National aggregate + model governance	Policy decisions on clinical protocols; readmission reduction targets
MOH_HMIS	Director of HMIS / ICT	Full national analytics + system config	Owns DHIS2 integration; TRIP platform governance at national level
MOH_ANALYST	National Health Data Analyst	National analytics + data export	Analyses national readmission trends; feeds MoH annual reports
MOH_PROGRAM	National Programme Officer (per disease)	Disease-specific national analytics	Malaria, HIV, TB programme officers see disease-stratified readmission data
TRIP_SYSADMIN	TRIP Platform System Administrator	Full system including config + audit	Manages all roles, integrations, model versions, and system health
AUDITOR	National Platform Auditor	Full read + audit export (no write)	Reviews immutable audit logs; compliance checks; NIMR reporting
ML_ENGINEER	ML / Data Engineer	Full data read + model management	Retraining, monitoring, SHAP validation, data pipeline governance
INT_OFFICER	Integration Officer	Integration config + sync management	Manages FHIR connectors, DHIS2 sync, Africa's Talking credentials

TIER 2 — ZONAL (6 Zones)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
ZONAL_COORD	Zonal Health Coordinator	Zonal aggregate analytics	Oversees all regions in zone; tracks zonal readmission burden
ZONAL_ANALYST	Zonal Data Analyst	Zonal analytics + facility ranking	Produces zonal health performance reports; feeds RHMT teams

TIER 3 — REGIONAL (31 Regions)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
RMO	Regional Medical Officer	Regional analytics + escalation review	Medical authority at regional level; reviews high-risk facility alerts
RHMT_LEAD	RHMT Chairperson	Regional analytics + dashboard full	Chairs Regional Health Management Team; performance accountability
RHMT_MEMBER	RHMT Member	Regional analytics (read)	Programme officers (MCH, Malaria, HIV) reviewing disease-specific rates
RHIM	Regional Health Information Manager	Regional analytics + facility approval	Manages regional HMIS; approves facility onboarding; data quality authority

TIER 4 — DISTRICT / COUNCIL (185 LGAs)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
DMO	District Medical Officer	District analytics + clinical escalation	Medical authority at district; reviews district readmission rate alerts
CHMT_LEAD	CHMT Chairperson	District analytics + dashboard full	Council Health Management Team lead; performance accountability
CHMT_MEMBER	CHMT Member	District analytics (read)	District programme officers reviewing their disease area data
DHIO	District Health Information Officer	District analytics + facility analytics + DHIS2 sync	Owns district HMIS data quality; approves new facility onboarding; primary TRIP liaison
DISTRICT_PHARMACIST	District Pharmacist / DIO	District drug-related readmission analytics	Reviews medication-related readmission patterns across district facilities

TIER 4 — FACILITY MANAGEMENT (Hospitals + Health Centres)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
FACILITY_DIRECTOR	Hospital Director / Medical Officer in Charge	Facility aggregate + staff management view	Overall facility performance; readmission rate accountability; staff role management
TFADMIN	TRIP Facility Administrator	Facility config + role assignment + integration status	Onboards facility to TRIP; assigns roles; monitors sync; manages local config
WARD_MANAGER	Ward Sister / Charge Nurse	Ward-level patient list + predictions	Manages clinical staff in the ward; ensures discharge workflow compliance

TIER 4 — CLINICAL STAFF (Patient-Facing)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
CLINICIAN	Medical Officer / Doctor	Own facility patients + full predictions + SHAP	Primary prediction consumer; acts on high-risk scores; completes discharge checklist
CLINICAL_OFFICER	Clinical Officer / AMO	Own facility patients + predictions + SHAP	Same as CLINICIAN — CO/AMOs are the majority of prescribing staff in Tanzania facilities
NURSE	Registered Nurse / Enrolled Nurse	Own ward patients + risk flags only	Monitors ward patients; escalates high-risk flags to clinician; CHW liaison
MIDWIFE	Midwife	Obstetric ward patients + maternal risk flags	Obstetric readmission risk (PPH, eclampsia, neonatal); antenatal flag review
PHARMACIST	Pharmacist	Medication-relevant patient data + drug readmission flags	Reviews high-risk medication lists; polypharmacy alerts; discharge medication reconciliation
LAB_TECHNICIAN	Laboratory Technician / Scientist	Lab-relevant patient flags (critical values)	Notified when critical lab values (low Hb, high creatinine) increase readmission risk score

RADIOGRAPHER	Radiographer	Imaging request flag only	Notified for urgent imaging requests triggered by high-risk patient workflow
NUTRITIONIST	Nutritionist / Dietitian	SAM and nutrition-flagged patients	Reviews severe acute malnutrition readmission flags; nutritional rehabilitation planning
SOCIAL_WORKER	Medical Social Worker	Social determinant fields + transport + follow-up	Reviews transport barriers, livesAlone flag, poverty indicators in readmission risk

TIER 4 — ADMINISTRATIVE AND RECORDS

Role Code	Role Title	TRIP Access Level	Key Responsibilities
HRO	Health Records Officer	Full patient record (admin) + data entry	Enters admission/discharge data; validates ICD-10 codes; manages patient pseudonyms
DATA_CLERK	Data Entry Clerk	Limited data entry (supervised by HRO)	Supports HRO in high-volume facilities; cannot view predictions
NHIF_OFFICER	NHIF / CHF Desk Officer	Insurance fields only	Verifies insurance coverage; updates NHIF/CHF status affecting social risk score

TIER 5 — HEALTH CENTRES + DISPENSARIES

Role Code	Role Title	TRIP Access Level	Key Responsibilities
HC_OFFICER	Health Centre Officer in Charge	Facility aggregate + patient list (simple view)	Manages health centre; reviews facility readmission rate; manages CHW assignments
HC_NURSE	Health Centre Nurse / Nurse in Charge	Own facility patients (simplified interface)	Simplified clinical dashboard for health centre context; referral flag visibility
HC_HRO	Health Centre Records Officer	Data entry + basic patient record	Paper-to-digital bridge; enters batch data from registers

TIER 6-7 — COMMUNITY (CHW / Village Level)

Role Code	Role Title	TRIP Access Level	Key Responsibilities
CHW	Community Health Worker	Assigned follow-up list only (mobile, offline-first)	Executes 7-day phone call and 30-day home visit follow-ups; records outcomes
VHW	Village Health Worker	Basic follow-up notification (SMS-driven)	Receives SMS alerts for high-risk discharged patients in their village; reports back via USSD
CHW_SUPERVISOR	CHW Supervisor / Ward Health Coordinator	CHW team follow-up dashboard	Monitors CHW follow-up completion rates; escalates missed follow-ups to HC officer

3. Role Permission Matrix

The following matrix maps each role group to data access scope and action permissions. This is the basis for RBAC implementation — every API endpoint must enforce these permissions using `requirePermission()` + facility scope middleware.

Role / Group	Patient PII	Predictions	Facility Aggr.	District Aggr.	National Aggr.	Configure
MOH_PS / MOH_DCS	None	None	None	None	Read	None
MOH_HMIS / MOH_ANALYST	Anonymised	Read	Read	Read	Full read	System config
TRIP_SYSADMIN	Technical only	Full	Full	Full	Full	Full
AUDITOR	Audit-gated	Read all	Read all	Read all	Read all	None
ML_ENGINEER	De-identified	Full + manage	Full	Full	Full	Model mgmt
ZONAL_COORD / ANALYST	None	None	Read zone	Read zone	None	None
RMO / RHMT_LEAD	None	None	Read region	Read region	None	None
RHIM	None	Read region	Read region	Read region	None	Region onboard
DMO / CHMT_LEAD	None	None	Read district	Read district	None	None
DHIO	None	Read district	Read district	Read district	None	District approval
FACILITY_DIRECTOR	None	Read facility	Read facility	None	None	None
TFADMIN	None	Read facility	Read facility	None	None	Facility config
CLINICIAN / CLINICAL_OFFICER	Own facility (PII)	Own facility full	Own facility	None	None	None
NURSE / MIDWIFE	Own ward (PII)	Own ward risk flags	None	None	None	None
PHARMACIST	Medication data only	Drug-risk flags	None	None	None	None
HRO / DATA_CLERK	Own facility (admin only)	None	None	None	None	None
CHW	Assigned patients only	Follow-up list	None	None	None	None
VHW	SMS-notified only	None	None	None	None	None

4. Dashboard Personas — What Each Role Sees on Login

Every role lands on a dashboard designed around their operational reality, not around system architecture. A clinician does not need to see DHIS2 sync status. A DHIO does not need individual patient names. The following defines the exact dashboard composition per persona.

National Intelligence Dashboard — MOH_HMIS / MOH_ANALYST / MOH_PS

MOH_NATIONAL — National Health Intelligence Dashboard *[National Tier]*

Data Access	Key Actions	Dashboard Focus
- National 30-day readmission rate (last 30 days, trend vs last year) - Readmission rate by region (all 31 regions) — choropleth map - Top 10 facilities by readmission burden nationwide - National disease breakdown: malaria, HIV, TB, SAM, sickle cell rates - Model performance panel: AUC drift status, calibration health - DHIS2 sync status (last sync timestamp)	- Export national readmission intelligence report (PDF/Excel) - Drill down to zone → region → district → facility - Set national readmission rate target (policy threshold) - Configure data retention and PII policy settings (SYSADMIN only) - Trigger DHIS2 national org unit sync (INT_OFFICER)	- Tanzania regional readmission map (choropleth, live data) 12-month national trend line chart - Top facilities table (sortable by rate, tier, region) - Disease-stratified donut chart - Model health panel (green/amber/red status indicators) - Alert feed: new drift detected, calibration bias flagged

Regional Intelligence Dashboard — RHMT_LEAD / RMO / RHIM

MOH_REGIONAL — Regional Health Management Dashboard *[Regional Tier]*

Data Access	Key Actions	Dashboard Focus
- Regional 30-day readmission rate vs national benchmark - All district readmission rates within their region - Facility ranking within region (highest burden first) - Disease breakdown by district - Pending facility onboarding approvals (RHIM)	- Approve new facility registrations (RHIM only) - Send alert to DMO/CHMT for high-burden districts - Export regional performance report - Drill down to district → facility	- Regional readmission rate KPI card (vs national avg) - District comparison bar chart (sortable) - Facility ranking table within region - Disease breakdown stacked bar by district - Onboarding approvals queue (RHIM only)

District Intelligence Dashboard — DHIO / CHMT / DMO

DISTRICT — District Health Management Dashboard *[District Tier]*

Data Access	Key Actions	Dashboard Focus
- District aggregate 30-day readmission rate - All facilities in district with individual rates - High-risk patient count across district (anonymised count only) - Disease breakdown across district - Monthly trend for district (12 months) - CHW follow-up completion rate by facility	- Approve new facility onboarding (DHIO) - Assign DHIO analyst to a specific facility - Export district readmission intelligence report - Flag a facility for RHMT review if rate is critical - Trigger DHIS2 sync for district org units	- District rate KPI card - Facility ranking table (all facilities in LGA) 12-month trend line chart - Disease breakdown (malaria dominant most months) - CHW follow-up completion rate widget - Pending onboarding approvals

Facility Administration Dashboard — FACILITY_DIRECTOR / TFADMIN

FACILITY_ADMIN — Facility Intelligence Dashboard *[Facility Tier]*

Data Access	Key Actions	Dashboard Focus
- Facility 30-day readmission rate (vs district and national benchmark) - Monthly trend (12 months) - Disease breakdown of readmissions within facility - Integration health: FHIR sync status, last sync time - User management: active users by role, role expiry dates - Prediction model accuracy for this facility (calibration ratio)	- Assign roles to clinical staff (TFADMIN) - Configure facility's FHIR integration (TFADMIN) - Trigger manual data sync - Review and resolve failed sync items - Deactivate a user account (TFADMIN) - Add/edit facility metadata (bed count, ward names)	- Facility rate KPI card vs benchmarks 12-month readmission trend - Disease breakdown donut chart - Integration status panel (FHIR, DHIS2) - Staff role management table - Facility calibration health indicator

Clinical Dashboard — CLINICIAN / CLINICAL_OFFICER

CLINICIAN_DASH — Clinical Discharge Intelligence Dashboard *[Clinical Tier]*

Data Access	Key Actions	Dashboard Focus
- Today's discharge list with readmission risk scores and tier badges - Full patient record + SHAP explanations for own facility patients - Discharge workflow checklist progress per patient - Own facility 30-day readmission rate (facility average)	- View patient risk score and top 3 SHAP drivers at discharge - Complete discharge checklist steps - Override model risk category with documented reason - Assign follow-up task to CHW - Create new patient encounter	- Today's discharges ranked by risk score (High first) - Discharge workflow progress per patient (checklist) - My task queue (pending interventions) - Facility 30-day rate KPI widget - Last 5 alerts dispatched - Patient risk score detail panel (click-through)

- Pending intervention tasks assigned to self - High-risk patient alert history (last 7 days)	Mark intervention as completed	
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Ward Dashboard — NURSE / MIDWIFE / WARD_MANAGER

WARD_DASH — Ward Patient Safety Dashboard *[Ward Tier]*

Data Access	Key Actions	Dashboard Focus
- Own ward patient list with risk tier badge (High/Medium/Low only — no raw score) - Discharge planning patients flagged for same-day discharge - High-risk patients currently admitted (flag, no SHAP detail) - Follow-up tasks for own ward patients - Ward readmission rate for last 30 days	- Escalate high-risk patient flag to clinician - Mark discharge checklist step as completed (assigned steps) - View CHW assignment for a patient's follow-up - Record vital sign observation (triggers re-scoring if critical) - Request social worker review for high-risk patient	- Ward patient list with risk tier badges - Discharge planning queue (status: planning / workflow / discharged) - High-risk patients panel (today) - My checklist tasks widget - Ward 30-day rate KPI (simple bar)

CHW Mobile Dashboard — CHW / CHW_SUPERVISOR

CHW_DASH — Community Follow-Up Dashboard *[Community Tier]*

Data Access	Key Actions	Dashboard Focus
- Assigned follow-up list (patients discharged from own facility, own ward/village) - Patient pseudonym + risk tier + follow-up type due (phone call / home visit) - Due dates and overdue count - Outcome history (previous follow-ups completed) - Offline access: entire follow-up list cached locally	- Mark follow-up as completed with outcome (stable/readmitted/deceased/unavailable) - Record brief notes on patient condition - Flag patient for urgent facility review if deteriorating - View patient's discharge summary (pseudonymised — no PII shown on device) - Receive SMS reminder for due follow-ups (VHW path: USSD-only)	- Today's due follow-ups (sorted by urgency) - Overdue follow-ups alert panel - My completed follow-ups this week - Offline sync status indicator - Simple outcome entry form (mobile-optimised, 2G-compatible)

HRO Data Entry Dashboard — HRO / DATA_CLERK

HRO_DASH — Health Records Dashboard *[Records Tier]*

Data Access	Key Actions	Dashboard Focus
• Full patient record (admin fields: demographics, admission, diagnosis, discharge) • Facility patient register (all active + discharged) • Pending data entry items (incomplete records) • ICD-10 code validation errors • Data completeness report for facility	• Create and update patient records (encrypted PII auto-applied) • Enter and validate ICD-10 diagnosis codes • Mark encounters as complete for prediction trigger • Batch import data from facility paper registers • Run data completeness check and export gaps report	• Pending records queue (incomplete) • ICD-10 validation error list • Today's admissions table • Data completeness score widget • Recent data entries by this user

5. System Collaboration Architecture — The AI Injection Model

TRIP is the intelligence layer injected into Tanzania's health stack. It does not own patient records, drug stock, HR data, or insurance records — those belong to the systems that already manage them. TRIP reads from them, computes intelligence, and writes back structured clinical decision support. The integration architecture is OpenHIE-compliant, FHIR R4-based, and designed to be switched from demo instances to production with environment variables only.

5.1 Tanzania Health System Integration Landscape

System	Owner / Operator	Integration Type	What TRIP Gets From It
DHIS2 (HMIS)	MoHCDGEC ICT / TAMISEMI	REST API (aggregate + tracker)	Org unit hierarchy (all facilities, zones, regions), aggregate disease burden denominators, immunisation rates, OPD attendance — used as facility registry and analytics denominators
GoT-HoMIS	MoHCDGEC / TAMISEMI (PHC)	FHIR R4 via OpenMRS FHIR2 module	Patient encounters, ICD-10 diagnoses, discharge dates, vital signs, medications — the primary patient clinical data source for dispensaries and health centres
Afya eHMS	MoHCDGEC (Regional hospitals)	FHIR R4 via OpenMRS FHIR2 module	Same as GoT-HoMIS but at regional level — higher data quality and volume; primary source for most readmission events at Tier 2-3 facilities
CTC2	NACP (HIV programme)	NACP data sharing API (or FHIR bridge)	HIV enrollment status, ART start date, viral load (last) — used as binary features hasHiv + onArt; HIV status NEVER displayed in any UI output
TImR (SantéIMS)	MoHCDGEC (Immunisation)	FHIR Immunization resource	Vaccination history — relevant for neonatal and paediatric readmission risk; confirms BCG, measles coverage for SAM patients
eLMIS / mSupply	MSD (Medical Stores Dept)	REST / FHIR MedicationDispense	Drug dispensing records at facility level — confirms which medications a patient received; validates high-risk medication count feature
NHIF / OpenIMIS	NHIF + CHF (insurance)	REST API / CoverageEligibility	Insurance type (NHIF/CHF/TIKA/none) — used as social determinant feature; also used for billing integration (future phase)
iHRIS / HRHIS	TAMISEMI / MoHCDGEC (HR)	REST read-only	Staff-to-patient ratios by facility and ward — used as facility-level feature: understaffed wards correlate with premature discharge and higher readmission
HFR (Facility Registry)	MoHCDGEC	REST API	Master facility list with MoH codes, GPS, tier, ownership — canonical source for facility registration and validation
NBTS (Blood Bank)	NBTS (Blood Service)	FHIR Observation	Blood product administration — relevant for anaemia-related readmission flags; haemoglobin feature validation
DISA (HIV Viral Load)	NPHL / MoH Lab	REST API	Viral load results for HIV-positive patients — most recent VL used as model feature for HIV readmission sub-model (future phase)

Africa's Talking	Commercial (SMS/USSD)	SMS/USSD API	Outbound: high-risk discharge alerts to clinicians; CHW follow-up reminders; VHW USSD portal for reporting follow-up outcomes without a smartphone
PACS / Radiology	Individual hospital systems	HL7 v2 or FHIR Observation	Imaging flags in high-risk workflow — radiology results that elevate readmission risk; future phase for Muhimbili/KCMC
Patient Portal (future)	TRIP (own)	FHIR Patient / Consent	Patient-facing discharge summary; consent management for data use; appointment booking
GOVERNMENT / Lipa Nchi	e-Government (digital ID)	NIDA API / FHIR Patient	National ID number as future UPI anchor; address verification for social determinants

5.2 Integration Architecture — How Each System Connects

TRIP uses a mediator-based architecture modelled on OpenHIE. Each source system has a thin mediator service (deployed at or near the facility) that pulls FHIR resources, maps them to TRIP's internal schema, and pushes de-identified records to TRIP's central intelligence layer over HTTPS. TRIP never writes directly to source EMRs — it only writes back structured decision support via FHIR CDS Hooks at the point of discharge.

Integration Pattern	Systems Using It	Direction	Implementation
FHIR R4 Mediator Pull	GoT-HoMIS, Afya eHMS, NBTS	Source → TRIP	OpenMRS FHIR2 module exposes Patient, Encounter, Condition, Observation resources. TRIP mediator (Node.js) polls hourly, maps, de-identifies, upserts to TRIP data lake.
DHIS2 REST API Sync	DHIS2 HMIS, TImR	DHIS2 → TRIP (org units + aggregate)	Weekly org unit sync for facility hierarchy. Monthly aggregate data pull for denominators. Config-driven: play.dhis2.org in demo, hmis.dhis2.go.tz in production. One env var change.
REST API Data Pull	CTC2, eLMIS, NHIF, DISA	Source → TRIP	NACP/MSD data sharing agreement required. Pull patient-level features (HIV status, drug dispensing, insurance type) on-demand when a prediction is requested. Cache 24 hours.
FHIR CDS Hooks (Write-back)	Afya eHMS, GoT-HoMIS	TRIP → EMR	TRIP responds to CDS Hooks calls from EMR at discharge event. Returns structured Card: risk score, top 3 drivers, intervention tasks. Clinician sees TRIP intelligence inside their familiar EMR screen.
SMS / USSD Gateway	Africa's Talking	TRIP → Clinician/CHW/VHW	Outbound SMS for high-risk discharge alerts, CHW follow-up reminders. Inbound USSD for VHW follow-up outcome reporting without smartphone. Tanzania operators supported: Vodacom, Airtel, Tigo, Halotel.
HFR REST Sync	MoH HFR	HFR → TRIP	Read-only facility registry sync. Weekly pull. New facilities in HFR trigger notification to DHIO for TRIP onboarding. Validates that every TRIP facility has a valid MoH facility code.
Offline Local Sync	PWA at facility (all facility tiers)	Local ↔ TRIP cloud	Service worker caches patient list and predictions. Sync queue stores new entries during offline periods. On reconnect, sync queue processes with retry logic. ONNX model for on-device inference during extended outages.

5.3 The CDS Hooks Pattern — Embedding TRIP in Existing EMR Workflows

The most clinically effective integration is not a separate TRIP tab that clinicians must navigate to — it is TRIP intelligence appearing inside the GoT-HoMIS or Afya eHMS discharge screen without any extra click. CDS Hooks is the HL7 standard that makes this possible. When a clinician sets a discharge date in the EMR, the EMR fires a hook to TRIP. TRIP responds within 2 seconds with a structured card: risk score, risk category, top 3 drivers in plain language, and a pre-built intervention plan. The clinician never leaves their familiar screen.

>> This is exactly how Epic's ReadmissionAdvisor works at UCSF and Stanford. It is not a popup or a separate portal. It is a blue card appearing in the EMR discharge screen saying: "TRIP readmission risk: HIGH (74%). Top drivers: 3 prior admissions in 12 months, active malaria, lives alone. Recommended: CHW

home visit at 7 days, clinic follow-up at 14 days." One click confirms. The prediction is logged. The task is created. The CHW is notified.

Hook Trigger	TRIP Response Card Content	Clinical Action Available
patient-discharge hook from Afya eHMS / GoT-HoMIS	Risk score + tier badge + top 3 SHAP drivers (human-readable labels, no HIV) + recommended follow-up plan	Confirm discharge with risk flag logged Override risk tier with reason Assign follow-up tasks
patient-view hook (admission)	Prior readmission alert if patient was discharged < 30 days ago from any facility	Acknowledge readmission event Add readmission flag to current encounter
medication-prescribe hook	High-risk medication alert if new prescription increases polypharmacy readmission risk	Accept prescription Request pharmacist review Modify prescription
order-sign hook (lab / radiology)	Flag if requested test suggests deteriorating condition elevating readmission risk	Acknowledge flag Escalate to senior clinician Update discharge plan

5.4 Data Flow Architecture — End to End

The following describes the complete data flow from patient admission to national intelligence, covering every integration touchpoint.

1. Patient admitted at any tier facility (dispensary to national hospital).
2. GoT-HoMIS or Afya eHMS creates a Patient resource and Encounter resource.
3. TRIP mediator polls the EMR every hour. Detects new Encounter. Pulls Patient (de-identified), Condition (ICD-10), Observation (vitals, labs), MedicationRequest (drugs).
4. TRIP checks CTC2 API for HIV/ART status (if patient consent collected or NIMR protocol covers). Updates hasHiv, onArt features.
5. TRIP checks NHIF/CHF API for insurance status. Updates insuranceType feature.
6. TRIP checks prior 30-day discharge history. If found: creates ReadmissionEvent, flags visit.readmittedWithin30d = true, alerts clinician.
7. At discharge: EMR fires CDS Hook to TRIP. TRIP runs predictionFeatureBuilder.js, calls ML service (or riskEngine.js fallback if offline). Returns structured CDS card within 2 seconds.
8. Clinician confirms. DischargeWorkflow created. Intervention tasks created if High risk. Africa's Talking SMS sent to CHW.
9. CHW receives SMS. Opens TRIP mobile app (offline-capable). Records 7-day and 30-day follow-up outcomes.
10. TRIP aggregates readmission events. Computes facility rates, district rates, regional rates, national rates.
11. TRIP pushes aggregate readmission indicators back to DHIS2 as custom data elements (optional but recommended). This makes TRIP intelligence visible in MoH's existing DHIS2 dashboards without requiring MoH staff to use TRIP directly.
12. Monthly: ML model monitoring checks AUC drift and facility calibration. Report generated and sent to ML_ENGINEER and MOH_HMIS.

5.5 What TRIP Gives Back to Each System

Interoperability is not one-directional. TRIP must write intelligence back into the systems it reads from, so that the entire Tanzania health stack becomes smarter — not just TRIP.

System	What TRIP Reads	What TRIP Writes Back
DHIS2	Org unit hierarchy, aggregate disease burden	Monthly aggregate readmission rate per facility as a DHIS2 data element — visible in MoH's own DHIS2 dashboards without requiring TRIP login
GoT-HoMIS (OpenMRS)	Patient encounters, ICD-10, vitals, meds	CDS Hook response: risk score card at discharge; optional FHIR Flag resource marking high-risk patients
Afya eHMS	Same as GoT-HoMIS at regional level	Same CDS Hook response; additionally a FHIR CarePlan resource with intervention tasks for structured EMR workflow
Africa's Talking	(Input: phone numbers from patient records)	SMS alerts to clinicians (high-risk discharge), CHW reminders (follow-up due), VHW USSD outcomes (follow-up completed/patient readmitted)
NHIF / OpenIMIS	Insurance coverage per patient	Readmission flag: if patient with NHIF is flagged High risk, notify NHIF desk officer so insurance pre-authorisation for readmission can be prepared in advance
iHRIS	Staff-to-patient ratios	Staffing alerts: if readmission rate spike correlates with low nurse-to-patient ratio in a facility, flag to DMO with both data points together

6. Problems This Platform Solves — Complete Problem Statement

Every design decision in TRIP — every role, every integration, every dashboard widget — exists to solve a specific, documented problem in Tanzania's health system. The following is the complete problem catalogue.

#	Problem	Who Feels It	How TRIP Solves It
1	No early warning at discharge: Clinicians discharge patients without knowing readmission risk.	Clinicians, nurses at all tiers	CDS Hook at discharge: risk score + SHAP drivers appear in EMR screen within 2 seconds of discharge date being set
2	70% of high-risk patients are missed: Current ML model has 30% recall — misses most readmissions.	ML engineers, MOH planners	IMPL-001 + IMPL-002: Tanzania features in fallback engine + synthetic data recalibration to 12% rate. Target recall > 45%.
3	Offline blindness: Fallback rules engine ignores malaria, HIV, TB, SAM — the dominant Tanzania readmission drivers.	Clinicians in rural/offline facilities	IMPL-001 wires Tanzania features into riskEngine.js. Fallback engine is now Tanzania-aware.
4	No readmission tracking: No 30-day prior discharge check; no ReadmissionEvent model; facility readmission rate is unknown.	DHIOs, CHMT, facility managers, MOH planners	IMPL-004: ReadmissionEvent model + prior discharge check on every new admission. Platform can now compute its own core metric.
5	Manual prediction trigger: Prediction must be manually requested; clinicians forget or skip it.	Clinicians, HROs	IMPL-005: Discharge workflow auto-triggers prediction the moment dischargeDate is set. No manual step required.
6	No post-discharge follow-up system: High-risk patients leave; no CHW is notified; no outcome is recorded.	CHWs, social workers, patients themselves	IMPL-007: FollowUpSchedule model + CHW mobile dashboard + SMS reminders + outcome recording
7	Privacy risk — HIV in SHAP: HIV status appears in clinical risk explanations shown on screens.	Patients, MoH privacy officers, NIMR reviewers	IMPL-003: SUPPRESSED_FEATURES set in predictor.py. HIV never appears in any TRIP output regardless of role.
8	Role fragmentation: 10 roles covering a 7-tier system with 60,000+ health workers is inadequate.	DHIO (doesn't exist), facility admins, auditors	IMPL-006 + this document: 35 roles with precise scope, delegated assignment, and expiry enforcement.
9	No national readmission intelligence: MoH has no visibility into 30-day readmission burden.	MoH policy makers, RHMT, World Bank/donor reporting	MOH national dashboard + DHIS2 write-back: readmission rate per facility appears in existing DHIS2 dashboards
10	Cross-facility readmission invisible: Patient discharged from Facility A, readmitted at Facility B — no alert.	DMOs, clinicians managing referral chains	Probabilistic matching (name+DOB+phone hash) flags likely cross-facility readmissions. UPI integration when Tanzania deploys it.
11	Data entry burden: HROs must enter data into multiple systems (GoT-HoMIS, DHIS2, facility registers).	HROs, data clerks	FHIR mediator pulls from GoT-HoMIS automatically. TRIP's DHIS2 write-back reduces manual re-entry.
12	Connectivity failure: Rural facilities lose internet for days; health workers cannot access clinical tools.	Clinicians and CHWs at Tier 5-7 facilities	Offline-first PWA + ONNX on-device inference + sync queue with retry. Full functionality at 2G or zero connectivity.
13	Uncalibrated model: Training on 24.6% readmission rate inflates all scores; clinicians lose trust in alerts.	All clinical roles	IMPL-002: fix synthetic data rate to 12%. Facility-level calibration layer (IMPL-011). Clinicians see reliable, well-calibrated scores.

14	No audit trail for data governance: MoH and NIMR cannot verify who accessed what patient data.	MoH ICT, NIMR ethics board, donor auditors	Immutable append-only AuditLog: every access, every prediction, every role change logged with timestamp, userId, and IP.
15	No CHW mobile experience: CHWs in Tanzania operate on feature phones or basic Android; no clinical app is designed for their context.	CHWs, VHWs, CHW supervisors	CHW dashboard: offline-first, 2G-optimised, Swahili labels, SMS/USSD fallback for VHWs without smartphones

7. Intelligence Expansion Roadmap

Readmission prediction is the entry point. Once the platform is trusted by clinicians and endorsed by MoH, the same infrastructure — FHIR mediators, role hierarchy, audit trail, ML pipeline, CHW follow-up chain — can host additional intelligence modules without rebuilding anything.

Module	Intelligence Type	Data Required	Target Deployment
Readmission Prevention (current)	30-day readmission risk at discharge	Patient clinical + social + facility data	Phase 0-1 (current work)
Sepsis Early Warning	Predict sepsis 4-6 hours before deterioration using vital sign trends	Real-time vital signs from Afya eHMS (streaming)	Phase 2 — requires real-time FHIR Observation feed
Malaria Severity Triage	Distinguish uncomplicated from severe malaria at triage to guide referral	Vital signs + RDT result + age + fever history	Phase 1 — can train on existing GoT-HoMIS malaria diagnosis data
Maternal Risk Intelligence	Predict PPH, eclampsia, maternal near-miss risk	Antenatal data from DHIS2 Tracker + delivery records	Phase 2 — requires ANC tracker integration
Neonatal Risk Stratification	Predict neonatal mortality and readmission risk in first 28 days	Birth weight, gestational age, APGAR, NICU stay	Phase 2 — paediatric wards at RRHs
Drug Supply Intelligence	Predict facility-level drug stockout risk from consumption trends	eLMIS/mSupply consumption data	Phase 2 — requires MSD eLMIS API agreement
Workforce Optimisation	Predict periods of critical understaffing correlating with readmission spikes	iHRIS leave records + admissions volume	Phase 3 — requires TAMISEMI iHRIS integration
NCD Progression Risk	Predict hypertension and diabetes complication risk for chronic care	NCD programme DHIS2 tracker + GoT-HoMIS chronic visit history	Phase 3 — requires NCD tracker expansion
Community Disease Surveillance	Aggregate CHW follow-up outcomes to detect local disease outbreak signals	CHW outcome reports + DHIS2 community data	Phase 3 — CHW follow-up module at scale

>> The business case to MoH: TRIP is not a readmission platform. It is Tanzania's national clinical intelligence infrastructure. Readmission is the proof of concept — the use case with the most immediately measurable value, the clearest data requirements, and the most direct impact on facility performance metrics that MoH already tracks. Once the infrastructure is trusted, every additional module costs a fraction of the first.

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